LLM COURSE PROJECT · PROPOSAL

ScamLoop-HE

Attacker-defender training for robust Hebrew SMS scam detection

Attacker LLM vs Defender classifier



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1 · MOTIVATING USE CASE

Hebrew scam SMS outpace the filters

Background

Israelis receive a steady stream of SMS impersonating the post, banks, tax authority and government services.

Why it matters

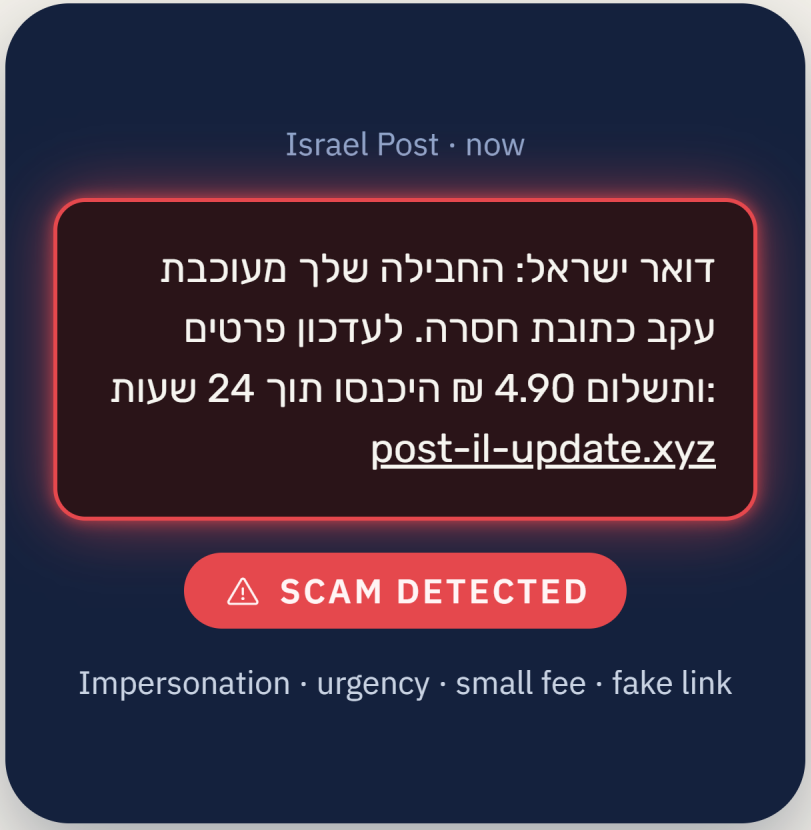
One click can lead to stolen card details and accounts, and a convincing message can fool anyone.

Why it’s hard

Little labeled Hebrew data, and scammers keep rewording to slip past filters.

Today

Generic spam filters and public warnings. We found no Hebrew scam dataset to train on.



2 · PROJECT TASK

Classify a Hebrew SMS - and stay robust

INPUT

One Hebrew SMS message (text, links, numbers)

OUTPUT

Scam / legitimate + scam type: delivery, bank, government, prize, other

RESEARCH QUESTION

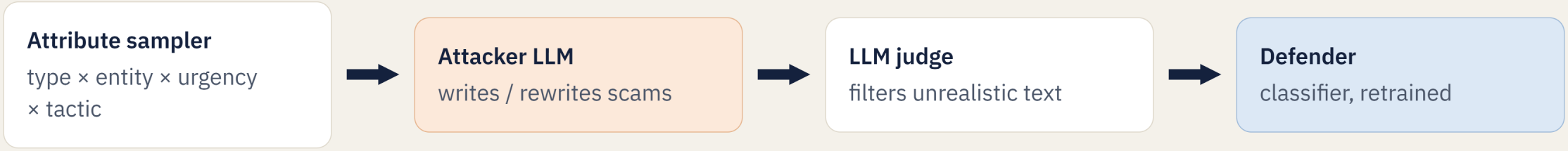
Does an iterative attacker-defender loop on native Hebrew data detect real and never-seen SMS phishing (smishing) campaigns better than training on translated English data?

Novelty 1. No public labeled Hebrew scam SMS set found; multilingual spam sets are machine-translated and skip Hebrew.

Novelty 2. Iterative loop that retrains the defender; prior work only attacks or trains once.

Novelty 3. Tested on a scam type held out from training, like a new campaign.

An attacker-defender loop



Each round: messages that **evade** the defender go back to the attacker for harder rewrites. The defender retrains on them plus a **replay buffer** of all earlier rounds, so it doesn't forget the simple scams.

Role	Models	Adjustment
Baseline (off-the-shelf)	GPT zero-/few-shot; classifier trained on translated English data	Prompting / translation
Fine-tuned defenders	DictaBERT / AlephBERT (Hebrew BERT models); SetFit (few-shot training)	Full fine-tune; few-shot contrastive
Attacker	Open LLM run locally (Ollama) + GPT via API	Prompted, feedback from defender

4 · DATA SPECIFICATION AND GENERATION

Label first, then generate

STEP 1 · Pick label + attributes

Label: scam
Type: delivery
Entity: Israel Post
Urgency: 24 hours
Tactic: fake short link

STEP 2 · LLM writes the SMS

דואר ישראל: החבילה שלך מעוכבת. לעדכון
כתובת תוך 24 שעות: bit.ly/il-post7

STEP 3 · Filter

LLM judge drops unrealistic messages; we review a sample by hand.

The label is chosen **before** the text, so every message is labeled automatically and nothing in it gives the answer away.

Synthetic scams
2,000 + adversarial rounds

Legitimate look-alikes
2,000, e.g. bank or HMO notices with official .co.il / .gov.il links

Real test set
150 real (75 scam + 75 legit), labeled by us

5 · METRICS AND KPIS

How we will know it works

Does it catch scams?	**Precision, recall, F1** on real messages and on a scam type never seen in training.
Does it survive new tricks?	**Evasion rate**: % of new attacker messages that slip through, each round. Should drop; loop stops below 5% or after 5 rounds.
Is our fake data realistic?	**LLM judge** scores realism; we check it against our own labels (Cohen's kappa).
Why did it decide that?	**SHAP** highlights the words behind each decision, especially the mistakes.

MAIN COMPARISON Translated English → Native Hebrew → **Native + attacker loop**